

Lesson 4 · Merit and Excellence Problem Solving

NCEA Level 2 Mathematics · Algebra

Handout

The one idea: general, complete, checked

At Excellence the algebra is the same as in Lessons 1–3. What changes is the **argument**: it must hold for **every** case (letters, not examples), find **every** answer, and be **checked** against the question, with a written conclusion. Markers write it as “correct mathematical statements” and “with reasoning”.

1 Algebraic proof

Words	Algebra
consecutive whole numbers	$n, n + 1, n + 2$
an even number	$2n$
an odd number	$2n + 1$
two different odd numbers	$2m + 1, 2n + 1$
a multiple of 3	$3k$
next day down a calendar	$n + 7$

n, m, k stand for whole numbers.

Example 1. Prove that the sum of three consecutive whole numbers is a multiple of 3.

$$n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1)$$

$3(n + 1)$ is 3 times a whole number, **so the sum is always a multiple of 3.** (r)

Example 2. Prove that $(2n + 1)^2 - (2n - 1)^2$ is a multiple of 8.

$$(4n^2 + 4n + 1) - (4n^2 - 4n + 1) = 8n$$

so it is always a multiple of 8.

Write it in this order

1. Name every number with letters.
 2. Write the expression the question is about, and simplify it.
 3. Take out the factor the question needs: $3(n + 1)$, $8n$, $2(\dots) + 1$.
 4. Finish with a sentence that says what the factor means: “so it is always a multiple of 3”.
- Examples, however many, are not a proof.

Example 3. On a calendar, a 3-by-3 square has top-left number n . Show that the diagonals have equal sums but never equal products.

n	$n + 1$	$n + 2$
$n + 7$	?	?
?	?	?

Corners: $n, n + 2, n + 14, n + 16$.

Sums: $n + (n + 16) = 2n + 16 = (n + 2) + (n + 14)$.

Products: $(n + 2)(n + 14) - n(n + 16) = 28$, never 0,

(r) so the products are never equal. (t)

2 Where a line meets a curve

Write it in this order

1. Make one variable the subject of the simpler equation ($y = \dots$ or $x^2 = \dots$).
2. Substitute into the other: you get a quadratic in one variable. Rearrange to $= 0$.
3. Solve, then find the **other coordinate for every value**. Reject values that give no real point.
4. Tangent (touches once): the quadratic has $b^2 - 4ac = 0$. Do not meet: < 0 .

Example 4. Where does $y = 2x - 3$ meet $y = x^2 - 2x - 3$? $x = 0: y = -3$; $x = 4: y = 5$. $(0, -3)$ and $(4, 5)$ (u)

$$x^2 - 2x - 3 = 2x - 3$$

$$x^2 - 4x = x(x - 4) = 0$$

Example 5. Find k so that $y = kx$ is a tangent to

$$y = x^2 + 3.$$

Example 6. Where do $x^2 + y^2 = 13$ and $y = x^2 - 1$ meet?

$$x^2 = y + 1, \text{ so } y + 1 + y^2 = 13:$$

$$x^2 - kx + 3 = 0, \quad k^2 - 12 = 0, \quad k = \pm 2\sqrt{3}$$

$$y^2 + y - 12 = (y + 4)(y - 3) = 0$$

$y = 3: x^2 = 4, x = \pm 2.$ $y = -4: x^2 = -3$, **no real point.**

(r) Points: (2, 3) and (-2, 3) **(t)**

3 Equations with a square root

Square both sides, solve the quadratic, then **check every answer in the original equation**: squaring can create an answer that does not work, because $\sqrt{\quad}$ is never negative. And $(x + 2)^2 = x^2 + 4x + 4$, not $x^2 + 4$.

Example 7. Solve $\sqrt{3x + 10} = x + 2$.

$$3x + 10 = x^2 + 4x + 4, \quad x^2 + x - 6 = (x + 3)(x - 2) = 0$$

Check $x = 2: \sqrt{16} = 4 = 2 + 2.$ Check $x = -3: \sqrt{1} = 1$ but $-3 + 2 = -1$, **rejected**. So $x = 2.$ **(r)**

4 Modelling and constraints

Example 8. A garden is 3 m longer than it is wide, with area 70 m². Find its size.

$$x(x + 3) = 70, \quad (x + 10)(x - 7) = 0$$

Example 9. A rectangle has perimeter 40 m. What is the largest area it can have?

$$\text{Sides } x \text{ and } 20 - x: x(20 - x) = A, \text{ so } x^2 - 20x + A = 0.$$

A width cannot be negative, so $x = 7$: **7 m by 10 m.**

A real width needs $400 - 4A \geq 0$, so $A \leq 100$: **at most 100 m², a 10 m square.** **(t)**

This is the move behind “find the largest value of a ”: write the condition as a quadratic, then ask when it has real solutions, then check the solutions against the context (a length must be positive).

5 Versions the marker won't accept

Question	Answer given	Why it falls short
Prove the sums of the corners are always equal	three squares checked by hand	Examples are not a proof. Use A for the corner and write every corner from it.
Find the co-ordinates of each intersection point	$x = \pm\sqrt{5}, x = \pm\sqrt{2}$	Co-ordinates need y too: four points, $(\pm\sqrt{5}, 2)$ and $(\pm\sqrt{2}, -1)$.
Largest whole-number a	$a = 7$, because $a^2 < 50$	The context also needs $p - a > 0$: with $a = 7$ that fails, so $a = 6$.

6 Ways to lose marks

From the wording of the marking schedules and from teaching observation; numbered for marking.

L22 Squaring a sum term by term. $(x + 3)^2 = x^2 + 6x + 9$.

L23 Solutions of a squared equation not checked. Put each answer back into the square-root equation.

L24 Proof by examples, or no conclusion. Letters for every case, then a sentence.

L25 Not every intersection point found. $x^2 = 4$ gives $x = 2$ and $x = -2$; each x needs its y .

L26 Solutions not checked against the context. Lengths and speeds are positive; sides like $p - a$ must be too.

7 Quick review

The discriminant (Lesson 3)

$$b^2 - 4ac > 0 \quad \text{two different real roots}$$

$$b^2 - 4ac = 0 \quad \text{one repeated root (tangent)}$$

$$b^2 - 4ac < 0 \quad \text{no real roots (no meeting)}$$

Squares to expand by sight

$$(x + 2)^2 \quad x^2 + 4x + 4$$

$$(x - 5)^2 \quad x^2 - 10x + 25$$

$$(2n + 1)^2 \quad 4n^2 + 4n + 1$$
